

```
In [224... import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import optimize as opt
```

```
In [226... data = pd.read_csv("data/velocities.csv")
```

```
In [228... print(data.head())
# Renaming the datas with their
```

	Trail	height(m)	height_err(m)	velocity(m/s)	velocity_err(m/s)
0	1	0.4	0.1	2.6	0.6
1	2	0.8	0.1	4.1	0.4
2	3	1.4	0.1	5.0	0.3
3	4	2.0	0.1	6.2	0.3
4	5	2.6	0.1	6.7	0.4

```
In [230... average_velocity = np.mean(data.iloc[:, 3])
print(f"The average velocity is {average_velocity:.2f} +/- {np.mean(data['velocity_
```

The average velocity is 5.87 +/- 0.34 m/s

```
In [234... data['veloctiy_squared(m/s)**2'] = data['velocity(m/s)']**2
data['veloctiy_squared(m/s)**2']
```

```
Out[234... 0    6.76
1   16.81
2   25.00
3   38.44
4   44.89
5   62.41
6   73.96
Name: veloctiy_squared(m/s)**2, dtype: float64
```

```
In [236... data.head()
```

```
Out[236...      Trail  height(m)  height_err(m)  velocity(m/s)  velocity_err(m/s)  veloctiy_squared(m/s)**2
0      1      0.4      0.1      2.6      0.6      6.76
1      2      0.8      0.1      4.1      0.4      16.81
2      3      1.4      0.1      5.0      0.3      25.00
3      4      2.0      0.1      6.2      0.3      38.44
4      5      2.6      0.1      6.7      0.4      44.89
```

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```
In [242... data['velocity_squared_err(m/s)**2'] = 2 * data['velocity(m/s)'] * data['velocity_e
```

```
In [246... data.head()
```

Out[246...

	Trail	height(m)	height_err(m)	velocity(m/s)	velocity_err(m/s)	veloctiy_squared(m/s)**2
0	1	0.4	0.1	2.6	0.6	6.76
1	2	0.8	0.1	4.1	0.4	16.81
2	3	1.4	0.1	5.0	0.3	25.00
3	4	2.0	0.1	6.2	0.3	38.44
4	5	2.6	0.1	6.7	0.4	44.89

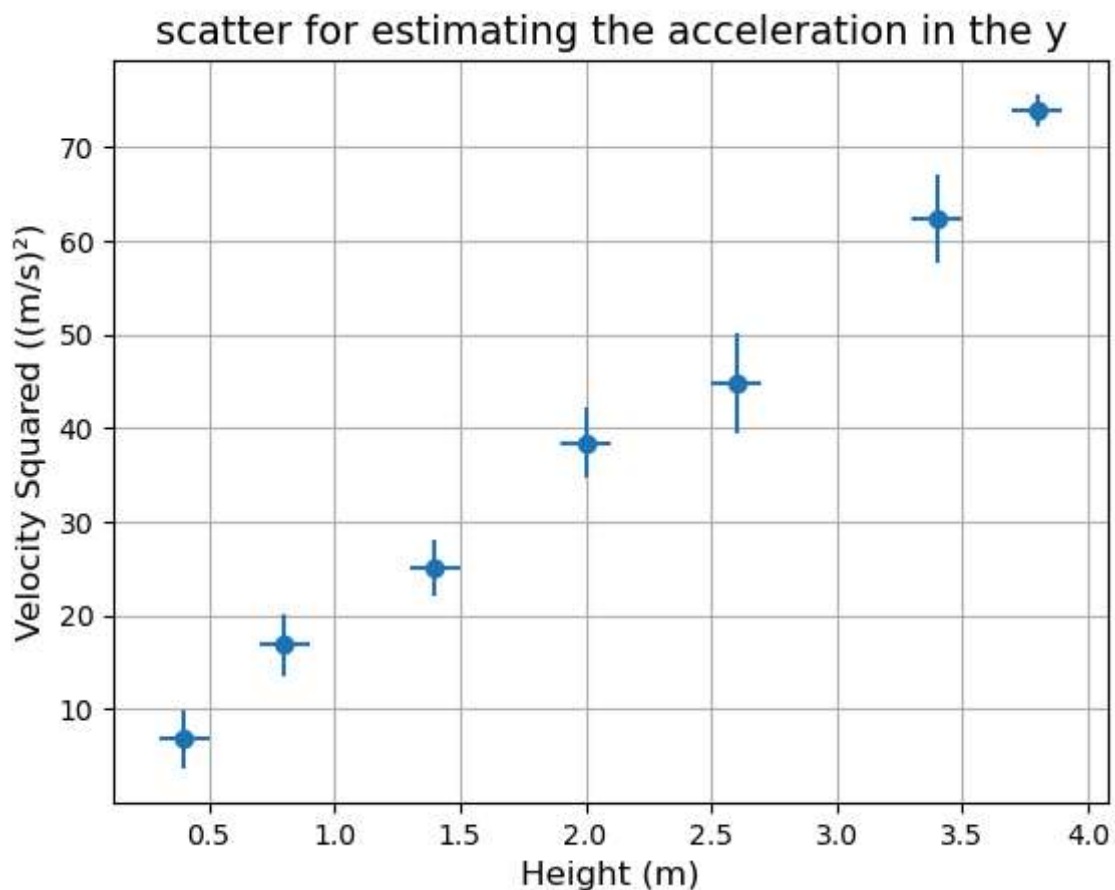
In [252...

```
print(data.columns.values)
```

```
['Trail' 'height(m)' 'height_err(m)' 'velocity(m/s)' 'velocity_err(m/s)'  
'veloctiy_squared(m/s)**2' 'velocity_squared_err(m/s)**2']
```

In [258...

```
x = data['height(m)']  
y = data['veloctiy_squared(m/s)**2']  
plt.errorbar(x, y, yerr=data['velocity_squared_err(m/s)**2'], xerr=data['height_err(m)'])  
  
plt.title("scatter for estimating the acceleration in the y", fontsize=14)  
plt.xlabel("Height (m)", fontsize=12)  
plt.ylabel("Velocity Squared ((m/s)^2)", fontsize=12)  
plt.grid(True)  
plt.show()
```



```
In [260...] def linear_model(x, A, B):
    return (A * x) + B
```

```
In [264...] par, cov = opt.curve_fit(linear_model,          # this function returns the paremet
    data["height(m)"], # this is the dependent variable (distan
    data["veloctiy_squared(m/s)**2"],          # this is a list of
    sigma=data["velocity_squared_err(m/s)**2"], # this is
    absolute_sigma = True)
```

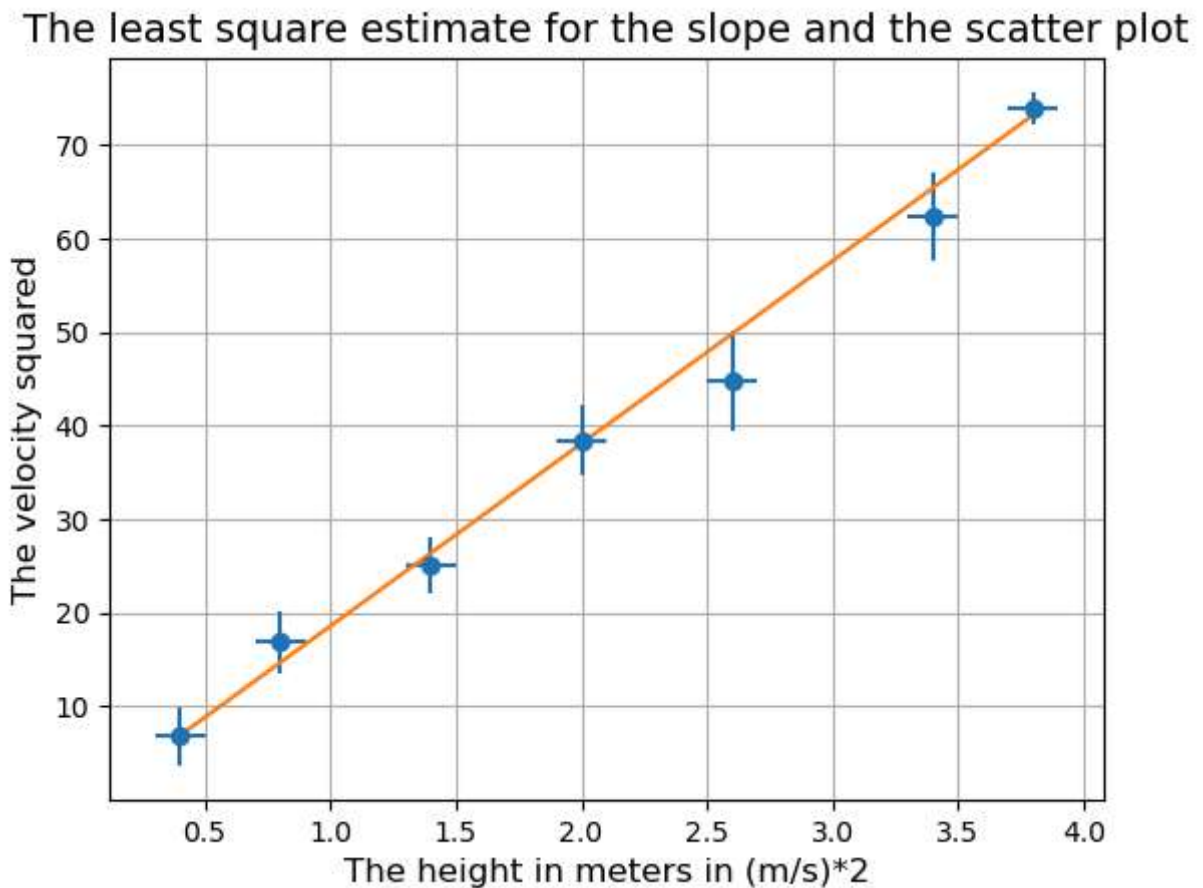
```
In [266...] print(par)
```

```
[19.53580107 -0.98268698]
```

```
In [274...] x = data['height(m)']
y = data['veloctiy_squared(m/s)**2']
plt.errorbar(x, y, yerr=data['velocity_squared_err(m/s)**2'], xerr=data['height_err

plt.title("The least square estimate for the slope and the scatter plot", fontsize=
plt.xlabel("Height (m)", fontsize=12)
plt.ylabel("Velocity Squared ((m/s)^2)", fontsize=12)
plt.grid(True)
x = np.linspace(min(data['height(m)']), max(data['height(m)']), 1000)
plt.xlabel("The height in meters in (m/s)*2")
plt.ylabel("The velocity squared")
plt.plot(x, x * par[0] + par[1])
```

```
Out[274...] [<matplotlib.lines.Line2D at 0x1a056546f10>]
```



```
In [276... std_err = [np.sqrt(cov[0, 0]), np.sqrt(cov[1, 1])]
```

```
In [278... print(f"2 * gravity = {par[0]} +/- {std_err[0]}")  
print(f"Gravity = {par[0] / 2:.2f} +/- {std_err[0]/2:.2f}")
```

```
2 * gravity = 19.535801065596154 +/- 0.8223302841524878  
Gravity = 9.77 +/- 0.41
```

```
In [280... print("My measured density: %2.1f +/- %2.1f m/(s**2)" % (par[0]/2, std_err[0]/2))
```

```
My measured density: 9.8 +/- 0.4 m/(s**2)
```

```
In [ ]:
```